

Weak Five Lecture Notes

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1 Polynomial Expressions

We are quite familiar with these already since you have taken calculus even if you do not realize it. A single *nomial* is a term inside any expression.

$$f(x) = x + 5 \tag{1}$$

In Eq (1) we can see this is a *monomial*. Lets break it down to the root words mono and nomial.

- mono: from the Greek word *monos*, meaning "alone".
- nomial: from the Greek word *nomós*, meaning "part of" or "portion".

We can discern the word now to mean some single piece. Tacitly referring to the existence of only a single term in Eq (1). This continues with *Binomial* and *Trinomial* but is eventually generalized to *Polynomial*. Where again the root word analysis will give you the meaning: many portions. Referring to the existence of multiple terms within the equation.

$$x + y + z \tag{2}$$

A polynomial expression can take the form of something like Eq (2), in which the equation has multiple terms. This concept is trivial to understand and happens often in different fields of mathematics. We now discuss the *coefficients* of polynomials. The coefficient can be root word analyzed though it may not be helpful. We will instead assert that the coefficient is any internal multiple of a term (what the hell does that mean????). This means that for Eq (2) the coefficients are $\{1, 1, 1\}$. Lets see a more common example.

$$5x^2 + 2x + 10 \tag{3}$$

We can see in Eq (3)poly2) a common polynomial to anyone who has studied calculus. The coefficients here are $\{5, 2, 10\}$. The reasoning here is to do with each term in the equation. The first term: $5x^2 \implies$ coefficient 5. The second term: $2x^1 \implies$ coefficient 2. The third term: $10x^0 \implies$ coefficient 10. The third term may seem a bit strange but it is true that any number (generally) raised to the power of zero is one, thus the logic is sound. Can we use this fact to encode specific kinds of polynomials into a computer

environment? Yes, If we know the structure of the polynomial then it is a trivial task to create a *coefficient row vector*.

$$5x^2 + 2x + 10 \implies \vec{v} = [5, 2, 10] \quad (4)$$

We have seen this before its just a row vector. In a programming environment we refer to this structure as a *coefficient array*, although for the purposes of this course array and row vector are synonyms. Matlab can find the roots of this polynomial for us automagically, using this kind of row vector.

```
vrow = [5,2,10]
roots(vrow)
```

The above code uses the MATLAB fxn *roots* to find the roots of the polynomial automagically. Likewise we can have MATLAB plot the expression with the *plot* fxn. MATLAB can help us do statistics on this kind of data as well. We are in an applied field thus we work in discrete space meaning we almost always have data before a fxn. MATLAB can help us analyze that data in order to fit a fxn to it. We can use *Interpolation*, where we fill in missing portions of data to help form a more continuous space. We can perform *Regression*, the prediction of data using what was already measured.

2 Interpolation

I will leave the VAST majority of this topic to your future stats professor. Generally interpolation is the estimation of missing data points using the surrounding data. Notice the estimation must be made on (dimensionally arbitrary) surrounded missing data. We can use the MATLAB fxn *interp1* for this purpose.

It is always a good idea to visualize any data throughout the process of interpolation.

3 Regression

I will leave the VAST majority of this topic to your future stats professor. Generally regression is the prediction of lone data from existing data. To make educated regressions we assume the data can be fit to a fxn, and there are no uncontrolled variables. We will almost always use the *polyfit* fxn for regression in this course.